

Chapter 7: Trigonometry & Periodic Functions

Radian Measure, Trigonometric Identities, Double Angles & Laws of Sines/Cosines

7.1 Fundamental Identities & Unit Circle

Angles are measured in radians: $\pi \text{ radians} = 180^\circ$. On the unit circle $x^2 + y^2 = 1$, $\cos(\theta) = x$ and $\sin(\theta) = y$.

Pythagorean Identities

- $\sin^2(\theta) + \cos^2(\theta) = 1$
- $1 + \tan^2(\theta) = \sec^2(\theta)$
- $1 + \cot^2(\theta) = \csc^2(\theta)$

Reciprocal & Quotient

- $\tan(\theta) = \sin(\theta) / \cos(\theta)$
- $\sec(\theta) = 1 / \cos(\theta)$
- $\csc(\theta) = 1 / \sin(\theta)$
- $\cot(\theta) = \cos(\theta) / \sin(\theta)$

7.2 Compound & Double Angle Formulas

Sum/Difference Formulas:

$$\sin(A \pm B) = \sin(A)\cos(B) \pm \cos(A)\sin(B)$$

$$\cos(A \pm B) = \cos(A)\cos(B) \mp \sin(A)\sin(B)$$

Double Angle Formulas:

$$\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$$

$$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta) = 2\cos^2(\theta) - 1 = 1 - 2\sin^2(\theta)$$

Worked Example 7.1: Identity Verification

Problem: Prove that $(1 - \cos(2\theta)) / \sin(2\theta) = \tan(\theta)$.

Solution: Replace $\cos(2\theta)$ with $1 - 2\sin^2(\theta)$ and $\sin(2\theta)$ with $2\sin(\theta)\cos(\theta)$:

$$[1 - (1 - 2\sin^2(\theta))] / [2\sin(\theta)\cos(\theta)] = [2\sin^2(\theta)] / [2\sin(\theta)\cos(\theta)]$$

Cancel terms: $\sin(\theta) / \cos(\theta) = \tan(\theta)$. (Q.E.D.)

7.3 Triangle Trigonometry: Sines & Cosines Laws

For any arbitrary triangle with side lengths a, b, c opposite angles A, B, C :

Law of Sines

$$a / \sin(A) = b / \sin(B) = c / \sin(C)$$

(Used when given ASA, AAS, or SSA)

Law of Cosines

$$c^2 = a^2 + b^2 - 2ab \cdot \cos(C)$$

(Used when given SAS or SSS)

Worked Example 7.2: Law of Cosines

Problem: In triangle ABC, $a = 5$, $b = 7$, $C = 60^\circ$. Find side c .

Solution:

$$c^2 = 5^2 + 7^2 - 2(5)(7)\cos(60^\circ)$$

$$c^2 = 25 + 49 - 70(0.5) = 74 - 35 = 39 \Rightarrow c = \sqrt{39} \approx 6.24.$$

Chapter 7 Review & Practice Problems

1. Convert 225° into radians and $5\pi/6$ radians into degrees.
2. If $\sin(\theta) = 3/5$ with θ in Quadrant II, evaluate $\cos(\theta)$, $\tan(\theta)$, and $\sin(2\theta)$.
3. Solve for $\theta \in [0, 2\pi)$: $2 \sin^2(\theta) - \sin(\theta) - 1 = 0$.
4. In triangle ABC, side $a = 8$, angle $A = 45^\circ$, angle $B = 60^\circ$. Find side b exactly.

